CSE STUDY HUB

Emerging Technology

UNIT 1 · Introduction to Emerging Technologies

What this unit is about

This unit builds understanding of **Introduction to Emerging Technologies**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Meaning, characteristics and drivers of emerging technologies

2. Technology evolution: major industrial revolutions and digital transformation

3. Industry 4.0: automation, connectivity, data and smart factories

4. Society 5.0: human-centred integration of physical and digital spaces

5. Converging technologies and multidisciplinary innovation

6. Adoption lifecycle, opportunities, risks and social impact

7. Ethics, sustainability and responsible innovation

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 1 NOTES · PART 1

Meaning, characteristics and drivers of emerging technologies

Start with a precise definition of Meaning, characteristics and drivers of emerging technologies. Explain its purpose, the problem it solves, its main components and where it is used. In an exam answer, add one practical example and distinguish it from the nearest related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Technology evolution: major industrial revolutions and digital transformation

Technology evolution: major industrial revolutions and digital transformation is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Industry 4.0: automation, connectivity, data and smart factories

For Industry 4.0: automation, connectivity, data and smart factories, identify the given values and the quantity to find before selecting a formula. Keep units consistent, show substitutions, calculate carefully and perform a reasonableness check. Write the formula and one solved example in your revision copy.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Introduction to Emerging Technologies, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 1 NOTES · PART 2

Society 5.0: human-centred integration of physical and digital spaces

Society 5.0: human-centred integration of physical and digital spaces is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Converging technologies and multidisciplinary innovation

Converging technologies and multidisciplinary innovation is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Introduction to Emerging Technologies, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 1 NOTES · PART 3

Adoption lifecycle, opportunities, risks and social impact

Present Adoption lifecycle, opportunities, risks and social impact as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Ethics, sustainability and responsible innovation

For Ethics, sustainability and responsible innovation, discuss the objective, stakeholders and possible harm. Cover consent/authority, confidentiality, integrity, accountability, transparency and applicable policy or law. Recommend safeguards and explain how they are checked.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Introduction to Emerging Technologies, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Meaning, characteristics and drivers of emerging technologies** — know the definition, core working and one example.

• **Technology evolution: major industrial revolutions and digital transformation** — know the definition, core working and one example.

• **Industry 4.0: automation, connectivity, data and smart factories** — know the definition, core working and one example.

• **Society 5.0: human-centred integration of physical and digital spaces** — know the definition, core working and one example.

• **Converging technologies and multidisciplinary innovation** — know the definition, core working and one example.

• **Adoption lifecycle, opportunities, risks and social impact** — know the definition, core working and one example.

• **Ethics, sustainability and responsible innovation** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Introduction to Emerging Technologies. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Meaning, characteristics and drivers of emerging technologies and explain its importance.

Q2. Explain the working/steps involved in Technology evolution: major industrial revolutions and digital transformation.

Q3. Differentiate between two related concepts from Introduction to Emerging Technologies.

Q4. Give one practical application of Ethics, sustainability and responsible innovation and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Introduction to Emerging Technologies.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.